Array & Matrix

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Agenda

1. Array
2. Matrix
3. Sawtooth Array
4. References

1

Arrray

Image in memory

index →	0	1	2	3	4
	5	7	2	4	9

Define array

```
DataType[ ]  variableName;
Or
DataType variableName [ ];
```

```
// Define int array
int[] arr1;
// Define long array
long[] arr2;
// Define float array
float[] arr3;
// Define double array
double[] arr4;
// Define boolean array
boolean[] arr5;
// Define string array
String[] arr6
```

Memory allocation

Option 1	`DataType[] variableName = new DataType [n] ;`
Option 2	`DataType[] variableName;` `variableName = new DataType[n];`

```java
// Define and allocation for int array
int[] arr1 = new int[5];// a.length = 5
// Define and allocation for long array
long[] arr2 = new long[5];// a.length = 5
// Define and allocation for float array
float[] arr3 = new float[7];//a.length = 7
// Define and allocation for double array
double[] arr4 = new double[7];//a.length = 7
// Define and allocation for boolean array
boolean[] arr5 = new boolean[8];//a.length = 8
// Define and allocation for string array
String[] arr6 = new String[6];//a.length = 6

```

Initialize

```
// Option 1
int[] arr = {1, 3, 5, 7, 9}; //a.length = 5
// Option 2
int [] arr = new int[5];
arr[0]=1;
arr[1]=3;
arr[2]=5;
arr[3]=7;
arr[4]=9;
```

Init

```
// Init long array
long[] arr1 = {1, 3, 5, 7, 9}; //a.length = 5
// Init float array
float[] arr2 = {1.3, 3.2, 5.5}; //a.length = 3
// Init double array
double[] arr2 = {2.3, 7.2, 9.5};//a.length= 3
// Init string array
String[] ngay = {
          "Sunday","Monthday","Tuesday",
          "Wednesday","Thurday","Friday","Staturday"
};//a.length= 7
```

foreach - statement

```
// variableName is item in array
for (DataType variableName : arrayName){
    Block code;
}
```

```
System.out.println("Number of items in array " + a.length);
for (int item : a){
    System.out.println(item);
}
```

```
System.out.println("Number of items in array " + a.length);
for (int i=0 ; i<a.length; i++){
    System.out.println(a[i]);
}
```

Input/output one-dimensional array

```java
// Input aray
System.out.print("Enter number of items: ");
int n = Integer.parseInt(scan.nextLine());
int [] a = new int [n]; //a.Length = n
for (int i = 0; i < a.length; i++) {
    System.out.print("a["+i+"]=");
    a[i] = Integer.parseInt(scan.nextLine());
}
// output use for
System.out.println("Output array with for");
System.out.println("Number of items is " + a.length);
for (int i = 0; i < a.length; i++)
{
    System.out.println(a[i]);
}
// output use foreach
System.out.println("Output array with foreach");
System.out.println("Number of items is " + a.length);
for (int pt : a)
{
    System.out.println(pt);
}
```

Sum all items in array

```
// use for
int s = 0;
for (int i = 0; i < a.length; i++)
{
     s = s + a[i];
}
System.out.println("s = {0}", s);

// use foreach
s = 0;
for (int pt : a)
{
     s = s + pt;
}
System.out.println("s = " + s);
```

Sort increase (Selection Sort)

```
int i, j;
int min, temp;
for (i = 0; i < a.length - 1; i++)
{
    min = i;
    for (j = i + 1; j < a.length; j++)
    {
        if (a[j] < a[min])
        {
            min = j;
        }
    }
    temp = a[i];
    a[i] = a[min];
    a[min] = temp;
}
```

Exercises

- Developing a program with:
  - Input one-dimensional array
  - Sum of items
  - Find MAX item
  - Sort increase
  - Print array to console

2

Matrix

Image in memory

Column index

Row index

	0	1	2	3	4
0	5	7	2	4	9
1	3	2	0	1	4
2	6	1	9	2	3

Define matrix

```
DataType [][]  variableName;
Or
DataType variableName [][];
```

```
// Define int matrix
int[][] arr1;
// Define long matrix
long[][] arr2;
// Define float matrix
float[][] arr3;
// Define double matrix
double[][] arr4;
// Define boolean matrix
boolean[][] arr5;
// Define String matrix
String[][] arr6;
```

Memory allocation

Option 1	DataType[][] matrixName = new DataType [n][m] ;
Option 2	DataType[][] matrixName; matrixName = new DataType [n][m];
n	Number of rows : matrixName.length
m	Number of columns : matrixName [i].length

```
// Define and allocation int matrix
int[][] arr1 = new int[3][5];
int numRow = a.length; //numRow = 3
int numColumn = a[i].length; //numColumn = 5
// Define and allocation long matrix
long[][] arr2 = new long[5][6];
int numRow = a.length; //numRow = 5
int numColumn = a[i].length;; //numColumn = 6
// Define and allocation float matrix
float[][] arr3 = new float[7][9];
int numRow = a.length; //numRow = 7
int numColumn = a[i].length; //numColumn = 9
```

Initialize

```
// Option 1
int[][] a = {
                { 1, 2 },
                { 3, 4 },
                { 5, 6 },
                { 7, 8 }
            };
int numRow = a.length; //numRow = 4
int numColumn = a[i].length; //numColumn = 2

// Option 2
int [][] a = new int[4][2];
int numRow = a.length; //numRow = 4
int numColumn = a[0].length; //numColumn = 2
int k = 1;
for (int i = 0; i < numRow; i++)
{
    for (int j = 0; j < numColumn; j++)
    {
        a[i][j]=k++;
    }
}
```

Input matrix

```java
// Input matrix
int[][] a;
System.out.print("Enter number of rows: ");
int n = Integer.parseInt(scan.nextLine());
System.out.print("Enter number of column: ");
int m = Integer.parseInt(scan.nextLine());
a=new int[n][m];
for (int i = 0; i < a.length; i++){
    for (int j = 0; j < a[i].length; j++){
        System.out.print("a[" + i + "][" + j + "] = ");
        a[i][j] = Integer.parseInt(scan.nextLine());
    }
}

```

Output matrix

```
// Output matrix
System.out.println("Number of rows :" + a.length);
System.out.println("Number of columns : {0}", a[0].length);
for (int i = 0; i < a.length; i++){
    for (int j = 0; j < a[i].length; j++) {
        System.out.print(a[i][j]+"\t");
    }
    System.out.println();
}

```

Sum all items in matrix

```
int s = 0;
for (int i = 0; i < a.length; i++){
    for (int j = 0; j < a[i].length; j++){
     s = s + a[i][j];
    }
}
System.out.println("s=" + s);
```

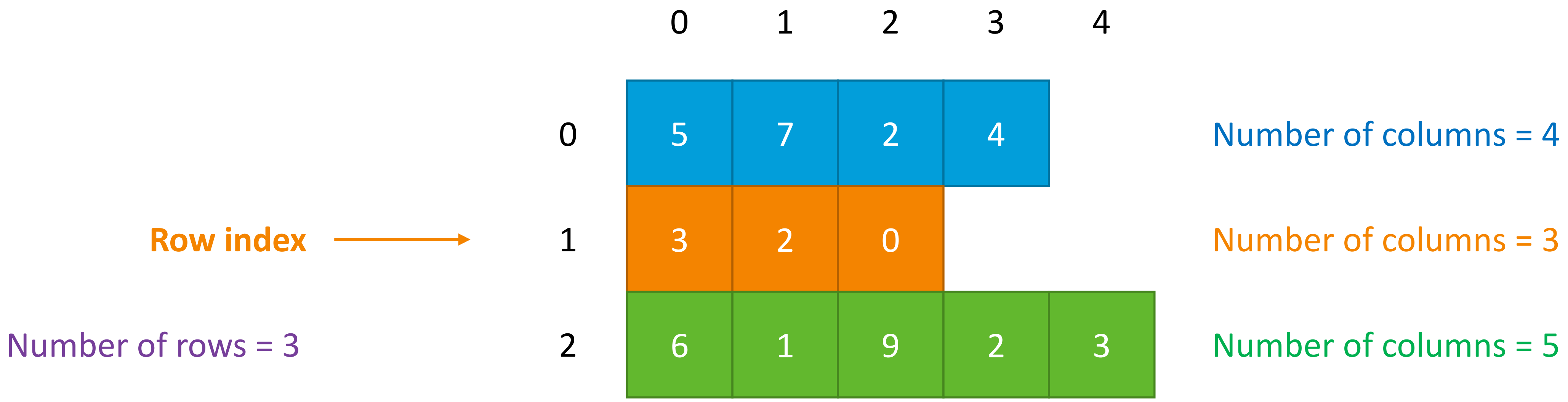
Exercises

- Developing a program with:
  - Input matrix
  - Sum of items in matrix
  - Find MAX items
  - Sort increase by Columns
  - Sort increase by Rows
  - Print matrix

3

Sawtooth Array

Image in memory



Define

```
DataType [][]  variableName;
Or
DataType variableName [][];
```

```
// Define int two-dimensional array
int[][] arr1;
// Define long two-dimensional array
long[][] arr2;
// Define float two-dimensional array
float[][] arr3;
// Define double two-dimensional array
double[][] arr4;
// Define boolean two-dimensional array
boolean[][] arr5;
// Define String two-dimensional array
String[][] arr6;
```

Memory allocation

Option 1	`DataType[][] arrayName = new DataType [n][] ;`
Option 2	`Datatype[][] arrayName;` `arrayName = new DataType [n][];`
n	Number of rows : arrayName.length

```java
// Define and allocate int sawtooth array
int[][] arr1 = new int[3][];
int numRow = arr1.length; //numRow = 3
// Define and allocate long sawtooth array
long[][] arr2 = new long[5][];
int numRow = arr2.length; //numRow = 5
// Define and allocate float sawtooth array
float[][] arr3 = new float[7][];
int numRow = arr3.length; //numRow = 7
```

Initialize

```java
int[][] a = {
                    { 1, 2 },
                    { 3, 4, 5 },
                    { 5, 6 , 7, 8 },
                    { 7, 8 ,9 ,10 }
               };
int numRow = a.length; //numRow = 4
int numColumn = a[i].length; //numColumn in row i
```

Input sawtooth array

```
// Input sawtooth array
int[][] a;
System.out.print("Enter number of row: ");
int n = Integer.parseInt(scan.nextLine());
a=new int[n][];
for (int i = 0; i < a.length; i++){
    System.out.print("Enter number of clumn at row " + i);
    int m = Integer.parseInt(scan.nextLine());
    a[i]=new int[m];
    for (int j = 0; j < a[i].length; j++) {
        System.out.print("a[" + i + "][" + j + "] = ");
        a[i][j] = Integer.parseInt(scan.nextLine());
    }
}
```

Output sawtooth array

```
// Output sawtooth array
System.out.println("Number of row is " + a.length);
for (int i = 0; i < a.length; i++){
    System.out.println("Number of column at row " + i + " is " + a[i].length);
    for (int j = 0; j < a[i].length; j++){
        System.out.print(a[i][j] + "\t");
    }
    System.out.println();
}
```

Sum all items

```java
int s = 0;
for (int i = 0; i < a.length; i++){
    for (int j = 0; j < a[i].length; j++){
     s = s + a[i][j];
    }
}
System.out.println("s = " + s);
```

Exercises

- Developing a program with:
  - Input sawtooth array
  - Sum of items in sawtooth array
  - Find row which have MAX sum of items
  - Print sawtooth array

4

References

References

- Books

  - Nguyễn Hoàng Anh, slide and video for Java programming course, FIT@HCMUS, 2010
  - The Java Language Specification Third Edition (2005)

- Online references

  - Java documentation: https://docs.oracle.com/en/java/


thank you
danke
謝謝
ngiyabonga
teşekkür ederim
gracias
merci
obrigado
bedankt
dziękuję
go raibh maith agat
mochchakkeram
sukriya
kop khun krap
terima kasih
спасибо
Баярлалаа
рахмат
dank je
tapadh leat
grazie
arigatô
dakujem
takk
taiku
sagolun
mauruuru
hvala
мерси
ありがとう
감사합니다
ευχαριστώ